

CO2 AT2 CSA1014 SOFTWARE ENGINEERING

Warehouse Robotics Coordination and Inventory Automation Platform

**PRESENTED BY
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1. Scenario & Problem Analysis

Scenario

An urgent order for 20 smartphones is received.

1. Manager views the order.
2. System checks stock.
3. Products are reserved.
4. A picking task is scheduled.
5. An available robot is assigned.
6. Robot picks the products.
7. Inventory is updated and the order is completed.

Problem

- Manual coordination causes delays.
- Stock information may become inaccurate.
- Robot allocation can be inefficient.
- Robot failure can interrupt tasks.
- Separate systems make monitoring difficult.

Need: a reliable architecture connecting users, services, databases and robots.

2. System Requirement Analysis

- Functional requirements: user authentication, order processing, stock checking and reservation.
- Task creation and prioritization, robot assignment, robot status monitoring.
- Inventory update after picking and order completion tracking.
- Quality requirements: scalability, reliability, performance, availability, security and maintainability.

3. Architecture Selection & Rationale

Selected Architecture

Layered Architecture

+

Service-Based Components

+

Deployment View

Separates presentation, business services, data and physical robot integration.

Why Suitable

- Separation of concerns
- Easier maintenance
- Independent service scaling
- Simple robot integration
- Better fault isolation
- Clear data ownership

Design Flow

Presentation



Application



Services



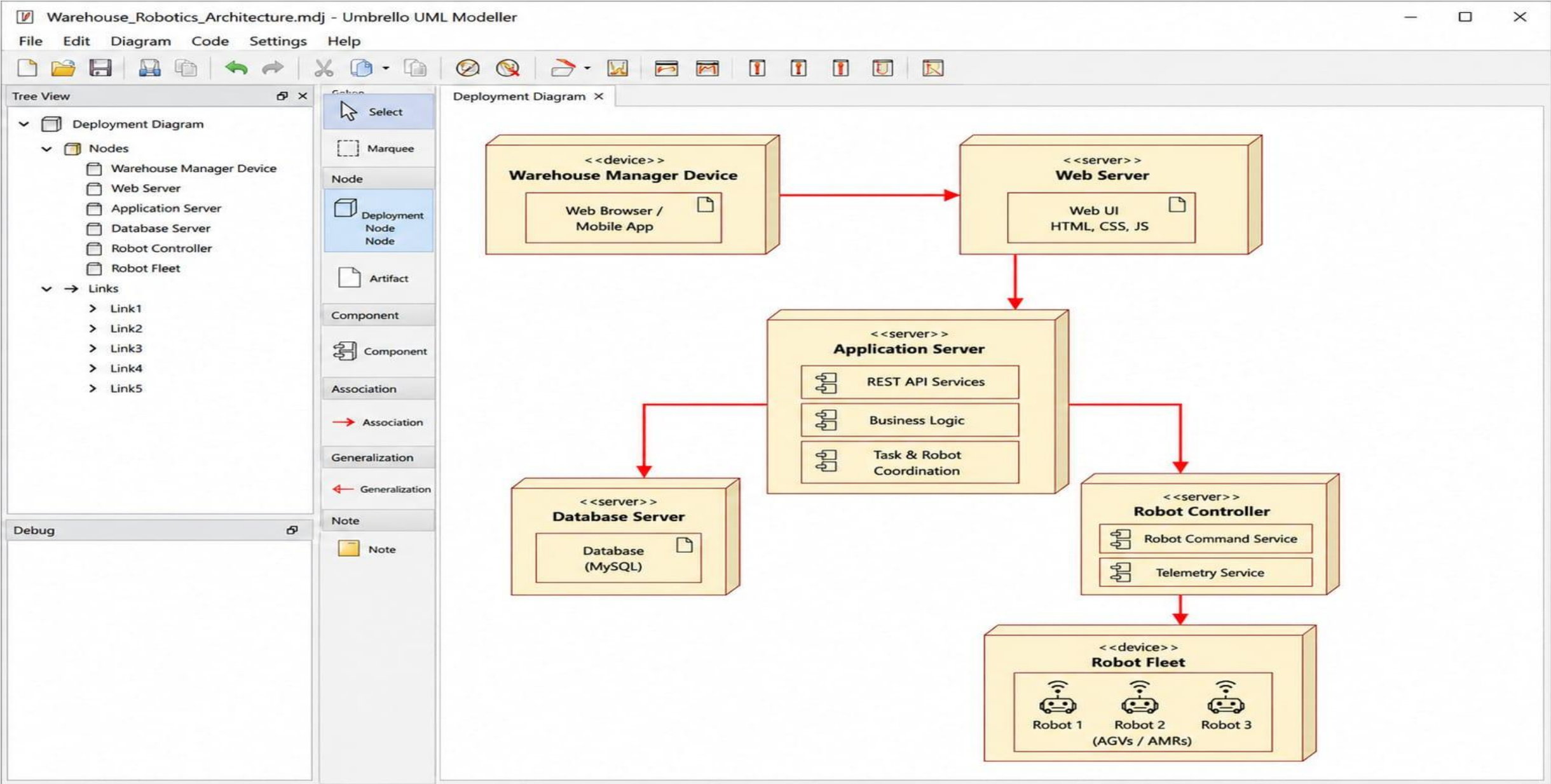
Data / Integration



Robot Environment

The deployment view maps these responsibilities to nodes.

4. Architecture Diagram



5. Deployment Nodes & Responsibilities

Manager Device

- Secure login
- Order monitoring
- Inventory view
- Robot/task monitoring

Uses the web dashboard to access the platform.

Application Server

- Order Service
- Inventory Service
- Task Scheduler
- Robot Service

Contains core business logic.

Data & Integration

- Warehouse Database
- Inventory Database
- Robot Controller

Stores data and connects software to hardware.

Robot Environment

- Robot 1
- Robot 2
- Robot 3
- Sensors/status

Executes picking tasks.

6. Scenario-Based Interaction Flow

1. Order

Manager views the urgent order.

2. Stock

Inventory Service checks availability.

3. Reserve

Required products are reserved.

4. Schedule

Task Scheduler creates a priority task.

5. Assign

Robot Service selects an available robot.

6. Execute

Robot Controller sends the picking task.

7. Update

Inventory is reduced after successful picking.

8. Complete

Order status changes to completed.

7. Interfaces & Communication

Software Interfaces

Web Dashboard → Application Services

Application Services → Database interfaces

Application Services → Robot Controller interface

Robot Controller → Robot communication channel

These interfaces reduce direct coupling.

Key Data Flows

Order Data → Order Service

Stock Data ↔ Inventory Service

Task Data → Task Scheduler

Robot Commands → Controller

Robot Status → Robot Service

Picking Result → Inventory Service

8. Quality Attributes & Design Support

Scalability

More robots can be added without redesigning the entire platform.

Services can scale independently.

Reliability

Robot failures can be isolated.

Central inventory data reduces inconsistent stock information.

Performance

Task scheduling and robot coordination are separated from presentation logic.

Security

Users access business services rather than directly accessing databases or robots.

9. Architectural Justification

- User devices are separated from business logic and data storage.
- Direct database access from users is avoided.
- Inventory, task scheduling and robot coordination can evolve independently.
- Additional robots and future warehouse locations can be integrated.
- Failures are easier to isolate and diagnose.
- The deployment model provides a practical blueprint for implementation in Umbrello.

10. Umbrello Implementation

Deployment Diagram

Create a Deployment Diagram in Umbrello.

Deployment Nodes:

- Manager Device
- Web/Application Server
- Database Server
- Robot Controller
- Robot Fleet

Artifacts / Components

- Web Dashboard
- Order Service
- Inventory Service
- Task Scheduler
- Robot Service
- Databases
- Controller Interface

Communication

- User access
- Service/API calls
- Database access
- Robot commands
- Robot status feedback

These relationships show deployment-time communication.

11. Expected Outcome & Result

- A clear deployment architecture for the warehouse automation platform is defined.
- Software responsibilities are mapped to appropriate deployment nodes.
- The scenario shows how an order moves through the deployed system.
- Robot control and inventory management are integrated without tightly coupling the user interface to hardware.
- The architecture provides a scalable, reliable and maintainable foundation for future implementation.

12. Conclusion

Conclusion

The proposed deployment architecture converts the warehouse scenario into a practical technical structure.

The Manager Device, Application Server, Databases and Robot Controller work together to process orders, coordinate robots and maintain accurate inventory.

Umbrello can be used to model and validate this architecture before coding.

Thank You